

Analysis of Federal Open Market Committee Communication: Topic Evolution and Sentiment Analysis (2018-2024)

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Abstract

This paper analyzes the communication of the Federal Open Market Committee (FOMC) from 2018 to 2024 using Natural Language Processing (NLP) techniques. We utilize BERTopic/clustering for topic modeling to identify three distinct phases of discourse: Monetary Policy, Inflation Shock, and Disinflation. Furthermore, we evaluate five sentiment analysis models, including general and domain-specific BERT variants, on a labeled dataset of FOMC documents. Our results demonstrate that the domain-specific FinBERT-FOMC model outperforms others (F1-score: 0.429), highlighting the importance of domain adaptation. We also identify a critical "Inflation Paradox" where the traditional sentiment-stance mapping inverts during periods of high inflation.

1 Introduction

Central bank communication has become a crucial tool for managing market expectations and guiding monetary policy. In recent years, the Federal Open Market Committee (FOMC) has increasingly relied on forward guidance to signal its policy intentions. However, the sheer volume and complexity of these communications make manual analysis challenging. Natural Language Processing (NLP) offers a scalable solution to quantify the stance and sentiment of these documents.¹

In this study, we apply NLP techniques to a corpus of FOMC meeting minutes, press conference transcripts, and speeches from 2018 to 2024. Our primary contributions are threefold: (1) we analyze the evolution of topics in FOMC communication using BERTopic; (2) we benchmark five sentiment analysis models, establishing the superiority of domain-specific models; and (3) we uncover a context-dependent relationship between sentiment

and policy stance, which we term the "Inflation Paradox."

2 Theoretical Background & Related Work

Effective analysis of Central Bank communication requires mapping sentiment to the "Taylor Rule" framework (Taylor, 1993): we interpret **Positive Sentiment** (confidence in growth) as a *Hawkish* propensity to tighten, and **Negative Sentiment** (recession risks) as a *Dovish* propensity to ease. While transformer models like **FinBERT** (Araci, 2019) outperform dictionary methods on financial news, they often struggle with the technical nuances of "Fedspeak." To address this, Gössi et al. (2023) developed **FinBERT-FOMC**, fine-tuning BERT specifically on decades of meeting minutes to capture domain-specific jargon. Shah et al. (2023) further argue that simple sentiment polarity often fails to capture policy intent, proposing expert-annotated datasets to explicitly distinguish between economic sentiment and monetary policy stance.

3 Dataset

3.1 FOMC Data

The primary dataset consists of official communications scraped from the Federal Reserve Board's website² (Jan 2018–Dec 2024), comprising three key document categories. **Meeting Minutes** provide detailed records of internal deliberations, including staff assessments and participants' policy views. While only four meetings were held in 2018, the frequency increased to eight per year from 2019. **Press Conference Transcripts** capture the Chair's post-meeting remarks, featuring both prepared statements and unscripted Q&A sessions that often reveal immediate policy signals; these also shifted from quarterly to meeting-by-meeting

¹Code and data processing scripts are available at: https://github.com/KTHxioayao/Fed_textmining_proj

²<https://www.federalreserve.gov/default.htm>

frequency in 2019. Finally, **Speeches** delivered by the Chair, Vice Chair, and Governors during inter-meeting periods offer focused perspectives on specific economic topics.

Data & Title Filtration. A custom ETL (Extract, Transform, Load) pipeline was engineered to process these unstructured PDF and HTML files. These pre-processing steps include geometric cropping to remove headers/footers from PDFs, and a rule-based speaker diarization algorithm to concentrate on Powell’s remarks from reporter inquiries in press conference transcripts. Another criterion in the filtering is that sentences not containing any keywords from the list (Shah et al., 2023) are removed, aiming to keep sentences to be meaningful for monetary policy inference. The final corpus consists of over 8202 sentence-level segments, structured with metadata for date, section (e.g., "Opening Statement" vs. "Q&A"), and document type. The detail of text data coverage and the filtering criteria are reported in Appendix A.

3.2 Labeled Data

The performance evaluation of our sentiment analysis models was performed on a stratified evaluation set including around 600 sentences, where classes were intentionally balanced (approx. 33% each for three classes: 0: Dovish, 1: Hawkish, and 2: Neutral). This manually annotated dataset is sourced from Shah et al. (2023), where sentences were sampled before categorization.

3.3 Economic Data

To validate the economic relevance of the extracted text sentiments, we collected monthly macroeconomic indicators from the Federal Reserve Economic Data (FRED) database³, specifically Consumer Price Index (CPI), Producer Price Index (PPI), and the Federal Funds Rate. Both CPI and PPI were transformed into Year-over-Year (YoY) percentage change rates using the formula $YoY_t = \frac{P_t - P_{t-12}}{P_{t-12}} \times 100$ to align with standard central bank inflation measurements.

4 Methods

4.1 Topic Modeling

We employ BERTopic (Grootendorst, 2022) to capture temporal shifts in FOMC communication. The pipeline first generates dense vector representations of text segments using the all-MiniLM-L6-v2

model. To ensure robust clustering, we apply UMAP (McInnes et al., 2018) for dimensionality reduction (n_components=5) and HDBSCAN (Campello et al., 2013) to identify dense regions (min_cluster_size=15). Finally, Class-based TF-IDF extracts representative keywords for each cluster, enabling the analysis of topic evolution shown in Figure 1.

4.2 Rule-based Model

A rule-based model works as a baseline in our study. A dictionary containing negative and positive words was created, and the occurrences of words in those two categories were counted for each sentence. A sentence is classified by the category that has more negative or positive words than the other. The construction of the dictionary is shown in Appendix B.

4.3 Pretrained Language Models

We evaluate four pretrained sentiment analysis models, leveraging **sequential transfer learning** to adapt general language representations to the financial domain. **FinBERT** (Araci, 2019) is a BERT-base model fine-tuned on financial news, while **FinBERT-FOMC** (Gössi et al., 2023) extends this with further fine-tuning on FOMC communications (2006-2022) to capture specific monetary policy jargon. For comparison, we include general-domain models: **RoBERTa-Large** (Liu et al., 2019), trained on 160GB of text, and **Cardiff-RoBERTa** (Barbieri et al., 2020), optimized for short-text social media sentiment. All models use their recommended classification heads mapped to our three-class schema (Hawkish/Neutral/Dovish).

4.4 Sentiment Index Construction

The sentiment index is calculated by aggregating the confidence scores of polarized sentences, normalized by the total count of hawkish and dovish sentences:

$$\text{Index} = \frac{\sum_{i \in H} S_i - \sum_{j \in D} S_j}{N_{\text{Hawkish}} + N_{\text{Dovish}}} \quad (1)$$

where S represents the model’s confidence score for the predicted class, and N represents the count of sentences in each category. Neutral sentences are excluded to focus on the intensity of polarized views.

³<https://fred.stlouisfed.org>

5 Results

5.1 Topic modeling

As shown in Figure 1, the discourse evolved through three distinct phases. Initially, the narrative alternated between Monetary Policy and Inflation. However, the landscape shifted following the 2020 rate cuts and the subsequent tightening cycle in 2022. During this period, the topic of 'Inflation & Down' surged, reflecting the Fed's urgent focus on price stability. As pressures eventually moderated, the discourse pivoted toward 'Percent & Inflation,' indicating a shift to technical disinflation targets. This trajectory demonstrates that FOMC communication is highly responsive to real-time economic realities.

5.2 Model evaluation & Comparison

Table 1 presents a comparative analysis of five distinct models evaluated on the FOMC dataset. The standard FinBERT model, which is trained on general financial text, underperforms its FOMC-tuned counterpart, highlighting the necessity of domain adaptation for this specific task. While other pre-trained language models (PLMs) generally outperform the rule-based baseline, a nuanced trade-off between accuracy and class balance is observed.

The highest raw accuracy (47.6%) is achieved by RoBERTa-Large model, while a closer inspection of its predictions reveals a significant bias toward the majority class: it predicts only 4.5% of sentences as non-neutral (Hawkish or Dovish). This suggests that while accurate overall, the model fails to capture the subtle polarity shifts critical for monetary policy analysis, effectively behaving as a sophisticated majority-class classifier.

In contrast, the domain-specific FinBERT-FOMC demonstrates superior capability in detecting sentiment polarity. Although its overall accuracy (42.9%) is slightly lower than that of RoBERTa-Large, it achieves the highest F1-score (0.429) and Macro F1 (0.38). Crucially, FinBERT-FOMC identifies a substantial proportion of non-neutral signals (53.0%), aligning more closely with the expected distribution of market-moving statements in central bank communications. This evaluation shows that FinBERT-FOMC model has the best balanced performance and it is then used to inference the sentiment from FOMC during year 2018 to 2024.

5.3 Sentiment Analysis and Economic Validation

Figure 2 illustrates a clear correlation between our sentiment metric and macroeconomic indicators. During the contractionary phase in 2020 when inflationary indices reached historical lows, the sentiment metric reached a deep bottom simultaneously, reflecting the Federal Reserve's strong dovish stance.

However, as the economy transitioned into the inflationary spike of 2021-2022, a clear divergence emerged. While surging risk-related terminology drove sentiment scores deep into negative territory, the Federal Reserve aggressively tightened policy. This divergence pitted the model's dovish signal against the Fed's actual hawkish policy, a contradiction we define as the 'Inflation Paradox' (analyzed in Section 6.2).

These findings suggest that while FinBERT-FOMC is a robust proxy for macroeconomic risk, its mapping to policy stance is regime-dependent, requiring careful interpretation during periods of supply-side inflation.

6 Discussion

6.1 Model Comparison & Error analysis

Out of 624 labeled sentences, 160 (25.6%) were misclassified by every model tested, underscoring the inherent difficulty of interpreting policy stance in financial texts. These consistent failures were most prevalent in FOMC Minutes (62 cases) and Press Conferences (62 cases), compared to Speeches (36 cases), suggesting that the formal and often ambiguous language of official records poses a greater challenge than direct oratory. The lower accuracies from those two categories are another evidence of this statement, as shown in Appendix C.

A significant observation is that models trained based on general BERT models without any specific fine-tuning on financial datasets have a comparatively low F1-score, revealing the nuance of sentiment classification in a financial context. Among those BERT-based models, FinBERT-FOMC, which was specifically fine-tuned on FOMC minutes, outperforms FinBERT in all metrics. Table 2 reveals this systemic failure. In this instance, specialized financial models erroneously predicted the stance as 'Dovish', whereas general-domain models (e.g., RoBERTa, Cardiff) defaulted to 'Neutral'.

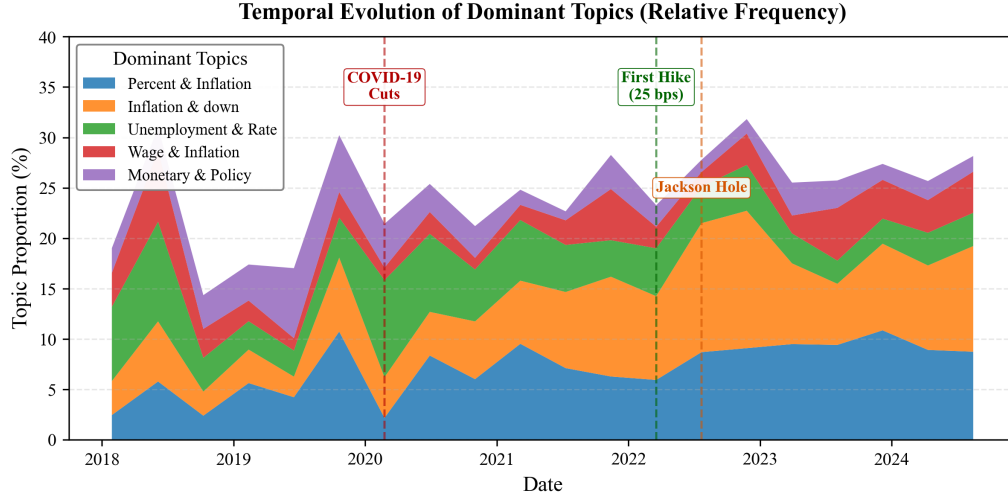


Figure 1: Evolution of 5 dominant topics, 2018-2024

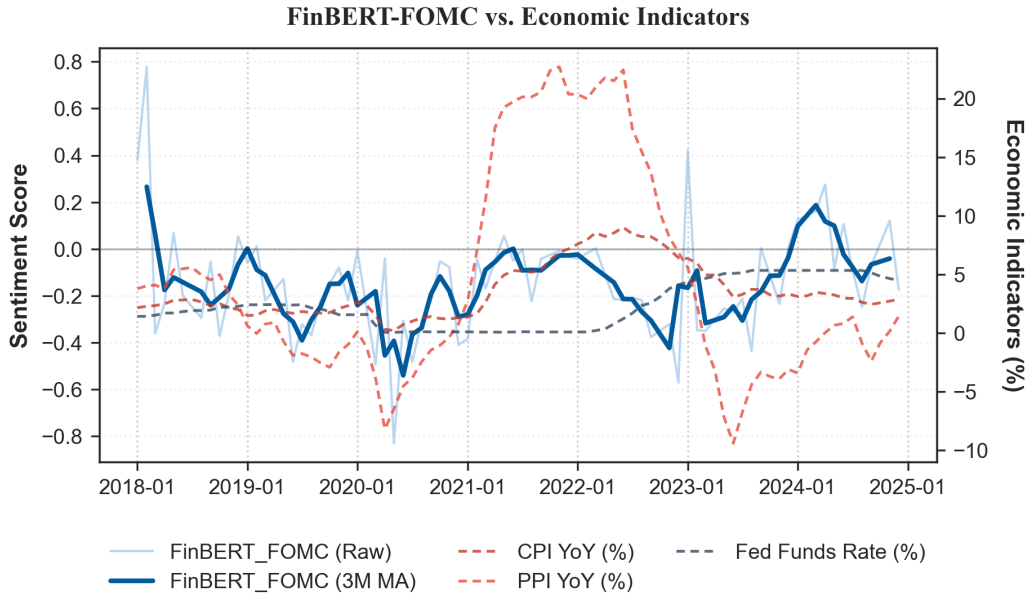


Figure 2: Performance Comparison of Sentiment Analysis Models on FOMC Dataset. Positive: Hawkish, negative: Dovish

Model	Accuracy	F1-Score	Domain	Non-neutral %	Macro F1
RoBERTa-Large	47.6%	0.328	General	4.5%	0.24
Cardiff-RoBERTa	43.4%	0.385	General	32.0%	0.33
FinBERT-FOMC	42.9%	0.429	Fed-Specific	53.0%	0.38
Baseline	39.3%	0.383	Rule-based	N/A	0.35
FinBERT	38.5%	0.403	Finance-General	45.2%	0.35

Table 1: Performance Comparison of Sentiment Analysis Models on FOMC Dataset. Dovish>0>Hawkish

In the first sentence presented in Table 2, models specialized for finance incorrectly predicted both stance as 'Dovish'. The error reveals a critical flaw in global sentiment mapping: financial models cor-

rectly identified 'inflation' and 'bringing down' as negative concepts. However, this mapping logic 'Negative = Dovish' fails in the specific context. When the Federal Reserve refer the history of com-

Text	Label	Base	FinBERT	FOMC	RoBERTa	Cardiff
During the 1980s and 1990s, the Federal Reserve succeeded in bringing inflation down from double-digit levels to the average rate of about 2 percent that has prevailed over the past decade.	Hawkish	Dovish	Dovish	Dovish	Neutral	Neutral
Nonetheless, employment is still 9.5 million below its pre-pandemic level for the economy as a whole.	Hawkish	Neutral	Dovish	Dovish	Neutral	Neutral

Table 2: Comparison of predictions. The source text is wrapped to a fixed width to allow space for all labels.

bating inflation, it implies the ambition of bring down inflation (Hawkish).

In the second given example, despite the presence of specific keywords ("employment," "below"), the sentiment is determined by adversative conjunctions ("Nonetheless," "still"), implying that handling conjunctions is critical for accuracy. This result is consistent with the findings of (Gössi et al., 2023) that utilizing Sentiment Focus (SF) method during pre-processing illustrated an improvement in accuracy by 5%.

6.2 The Inflation Paradox: Topic Dynamics and Sentiment Inversion

Our analysis identifies a critical regime shift in the relationship between sentiment and policy stance, heavily influenced by the evolution of dominant topics shown in Figure 1.

The Validity of Mapping (Pre–2021): Prior to 2021, the negative sentiment scores aligned well with the "Negative = Dovish" mapping. During this period, negative sentiment was primarily driven by concerns over growth, unemployment, and the inflation outlook (e.g., the COVID-19 shock), which correctly signaled the need for monetary accommodation.

Inflation Paradox (2021–2023): We identify this period as the core of the "Inflation Paradox." As inflation accelerated, the topic 'Inflation & Down' (Orange), likely capturing discussions on inflationary risks and downside economic pressure surged in topic proportion. This influx of risk-related terminology drove the FinBERT-FOMC sentiment score deep into negative territory. However, unlike previous cycles, this "negativity" was a trigger for aggressive tightening. Consequently, the mapping was distorted: while the model correctly detected economic distress (high inflation), the traditional proxy implied a 'Dovish' signal, contradicting the actual Hawkish policy response. This limitation highlights why alternative approaches,

such as those by (Shah et al., 2023), opt to bypass sentiment proxies by directly classifying text as hawkish or dovish. While their method achieves over 70% accuracy, it necessitates extensive annotation by domain experts, a resource-intensive trade-off.

The Stabilization Turn (2023– Present): The year 2023 marks a pivotal inflection point. As the discussion shifted toward 'Percent & Inflation' (Blue), reflecting technical discourse on disinflation, the panic associated with the initial inflation shock (Orange) subsided. Consequently, sentiment scores began to climb. This recovery represents a re-alignment of our mapping: the rising sentiment scores once again are consistent with the sustained Hawkish policy stance.

Conclusion

This study demonstrates that general-purpose language models are insufficient for analyzing central bank communication. Our benchmarking shows that domain adaptation is not merely beneficial but necessary to capture the technical nuances of monetary policy.

Most importantly, we identify a "regime-dependent" failure in traditional sentiment analysis. During the 2021-2023 inflation shock, the standard mapping of "negative sentiment equals dovish stance" collapsed, as the Federal Reserve used negative economic descriptions to justify aggressive tightening. This "Inflation Paradox" indicates that static sentiment models are prone to misinterpreting signals during economic turning points. Future research should therefore prioritize dynamic models that can adjust sentiment-stance mappings based on the prevailing macroeconomic environment.

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A Text data and filtering criteria

Source	Sentence Count	Word Mean	Word Std	Word Min	25%	50%	75%	Max
Minutes	1047	25.41	11.16	5	17	23	33	80
Press Conf	5040	14.44	7.47	5	9	13	18	64
Speech	2115	21.82	8.91	5	15	21	28	73

Table 3: Descriptive statistics of sentence lengths (number of words) across data sources.

Filtering Keywords. The filtering process retains sentences containing at least one of the following domain-specific keywords: *inflation expectation, interest rate, bank rate, fund rate, price, economic activity, inflation, employment, unemployment, growth, exchange rate, productivity, deficit, demand, job, market, monetary policy*.

B Rule-based Model Dictionary

The following lists detail the keywords used for the rule-based sentiment analysis baseline.

Positive Words: strong, growth, improvement, gain, solid, recovery, stable, progress, positive, confident, robust, healthy, sound, accommodative, stimulative, expansionary, easing, dovish.

Negative Words: weak, recession, decline, loss, difficult, negative, risk, inflation, tight, slow, uncertainty, deterioration, contraction, restrictive, hawkish, tightening, crisis, volatility.

C Detailed Model Performance by Source

Source	Sample Size	Baseline		FinBERT		FinBERT-FOMC		RoBERTa-Large		Cardiff-RoBERTa	
		Acc	F1	Acc	F1	Acc	F1	Acc	F1	Acc	F1
Speech	206	0.5146	0.4823	0.5194	0.4876	0.5243	0.4918	0.5728	0.5432	0.5340	0.5014
Minutes	227	0.3040	0.2715	0.2555	0.2234	0.3524	0.3187	0.4405	0.4123	0.3965	0.3689
Press	191	0.3665	0.3342	0.3927	0.3598	0.4188	0.3856	0.4136	0.3812	0.3717	0.3398

Table 4: Model accuracy and F1-score comparison across different FOMC data sources. Acc = Accuracy, F1 = F1-score. All values are weighted averages.